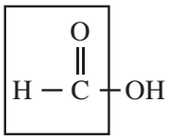
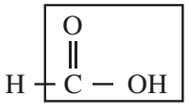
iv) Benzoic acid does not undergo friedal craft's reaction. This is due to the strong deactivating nature of the carboxyl group.

E) Reducing action of Formic acid

Formic acid contains both an aldehyde as well as an acid group. Hence, like other aldehydes, formic acid can easily be oxidised and therefore acts as a strong reducing agent



Aldehyde group



Carboxylic acid group

i) Formic acid reduces Tollens reagent (ammonical silver nitrate solution) to metallic silver.

$$HCOO^- + 2Ag^+ + 3OH^- \longrightarrow 2\,Ag + CO_3^{2-} + 2H_2O$$

(Tollens reagent) — Silver mirror

ii) Formic acid reduces Fehling's solution. It reduces blue coloured cupric ions to red coloured cuprous ions.

$$HCOO^- + 2Cu^{2+} + 5\,OH^- \longrightarrow Cu_2O + CO_3^{2-} + 3\,H_2O$$

(Fehlings solution) — red precipitate

Tests for carboxylic acid group

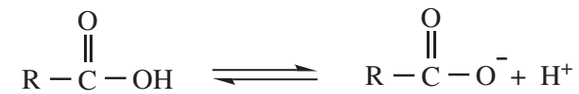
i) In aqueous solution carboxylic acid turn blue litmus red.

ii) Carboxylic acids give brisk effervescence with sodium bicarbonate due to the evolution of carbon-di -oxide.

iii) When carboxylic acid is warmed with alcohol and Con H_2SO_4 it forms an ester, which is detected by its fruity odour.

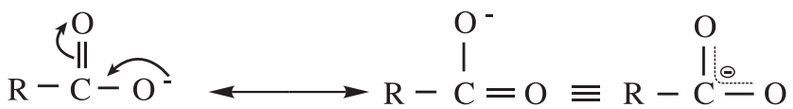
12.13 Acidity of Carboxylic acids

Carboxylic acids undergo ionisation to produce H^+ and carboxylate ions in aqueous solution. The carboxylate anion is stabilised by resonance which make the Carboxylic acid to donate the proton easily.



Carboxylic acid — Carboxylate ion

The resonance structure of carboxylate ion are given below.



The strength of carboxylic acid can be expressed in terms of the dissociation constant(Ka):

XII U12-Carbonyl Compounds.indd 177 11-01-2021 16:10:44